

Trainings ▾

General Information

Several different fan hub mounting locations and fan hub options are available. The illustrations in this procedure are generic and may **not** represent the hardware on all engines.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the accessory drive pulley belt, if equipped. See equipment manufacturer service information.
- Remove the cooling fan drive belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/220/220-008-002-tr.html>)
- Remove the electric fan clutch. Refer to Procedure 008-026 in Section 8 (</qs3/pubsys2/xml/en/procedures/216/216-008-026-tr.html>) or the equipment manufacturer service information.
- Remove the cooling fan, if equipped. See equipment manufacturer service information.
- Remove the accessory drive pulley, if equipped. Refer to Procedure 009-004 in Section 9. (</qs3/pubsys2/xml/en/procedures/216/216-009-004-tr.html>)
- Remove the fan spacer and pulley. Refer to Procedure 008-039 in Section 8. (</qs3/pubsys2/xml/en/procedures/216/216-008-039-tr.html>)

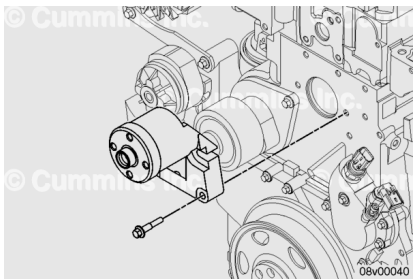
Remove

Note : There are many available fan hub configurations. Be sure to note the location, orientation, and mounting

pattern of the hub prior to removal from the engine.

Remove the fan hub mounting capscrews.

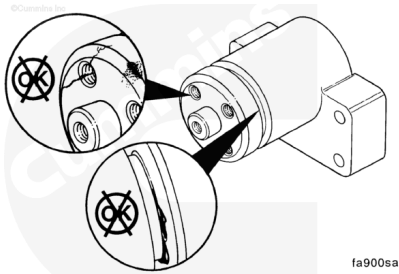
Remove the fan hub.



Initial Check

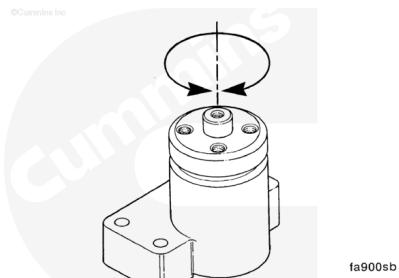
Inspect the fan hub for indication of oil seal leakage.

Inspect the fan hub for cracks and other damage.



Rotate the fan hub by hand to check for freedom of rotation.

The fan hub **must** rotate without any wobble or excessive end clearance.

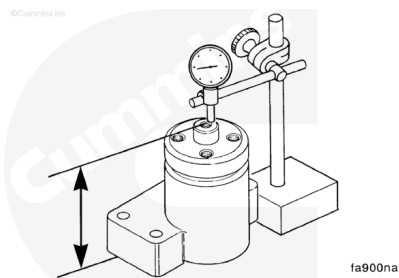


Measure the fan hub end clearance.

Fan Hub Shaft End Clearance		
mm		in
0.15	MAX	0.006

If damaged parts are found, the shaft does **not** rotate freely, or the shaft end clearance is **not** within specifications, the fan hub **must** be rebuilt or replaced.

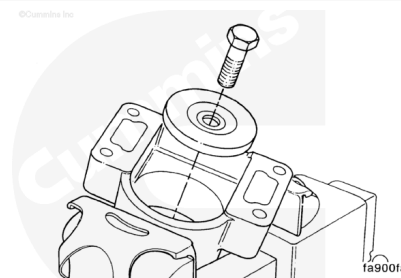
Note : Some fan hub assemblies can be rebuilt with new bearings and related components. Check for part availability before disassembling the fan hub.



Disassemble

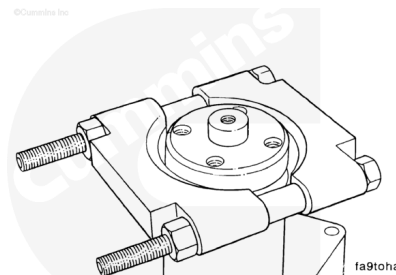
Secure the fan hub.

Remove the center bolt and retainer.

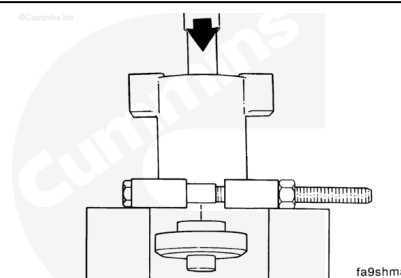


Install a bearing splitter as illustrated.

Note : Do not overtighten the bearing splitter as it may damage the shaft when the bearing is pressed out.

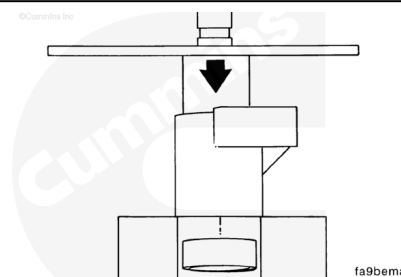


Support the fan hub housing/bearing splitter and press out the shaft.



Support the fan hub housing and press out the bearing.

Remove the snap rings.



Inspect for Reuse

Inspect the fan hub for damage to the:

- Snap rings
- Housing
- Shaft

- Bolt.

If damage is found, the fan hub can **not** be rebuilt and **must** be replaced as an assembly.

Assemble

⚠ CAUTION ⚠

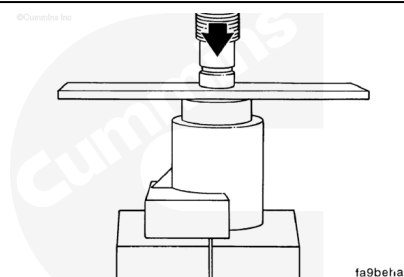
**Do not press the bearing by the inner portion of the bearing.
Damage to the bearing can result.**

Press a new bearing into the fan hub housing until it is even with the front of the housing.

Fan Hub Bearing Installation Force		
n		lbf
9000	MIN	2023

Replace the fan hub if the installation force is less than specified.

Install the snap rings.



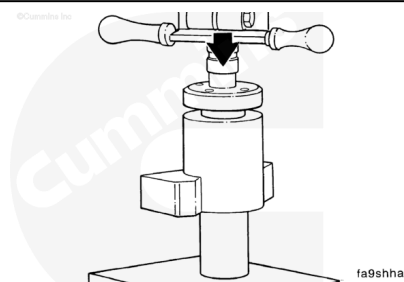
⚠ CAUTION ⚠

**Verify that the inner race is adequately supported to avoid
damage to the bearing. Do not press on the fan hub.**

Support the bottom of the inner race. Press the shaft into the bearing until it is seated.

Measure the assembled height. If the height is incorrect, disassemble and assemble with a new bearing.

Fan Hub Assembled Height		
mm		in
124.85	MIN	4.915
125.65	MAX	4.946

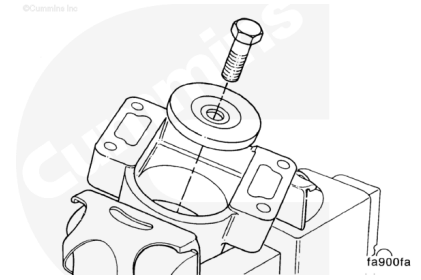


Secure the fan hub assembly.

Install the retainer and center bolt.

Tighten the center bolt.

Torque Value: 77 n•m [57 ft-lb]

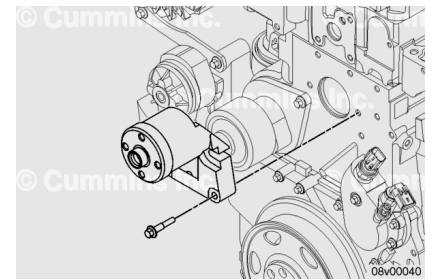


Install

Install the fan hub and mounting capscrews.

Tighten the capscrews.

Torque Value: 32 n•m [24 ft-lb]



Finishing Steps

⚠ WARNING ⚠

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- Install the fan spacer and pulley. Refer to Procedure 008-039 in Section 8.
(/qs3/pubsys2/xml/en/procedures/216/216-008-039-tr.html)
- Install the accessory drive pulley, if equipped. Refer to Procedure 009-004 in Section 9.
(/qs3/pubsys2/xml/en/procedures/216/216-009-004-tr.html)
- Install the electric fan clutch. Refer to Procedure 008-026 in Section 8
(/qs3/pubsys2/xml/en/procedures/216/216-008-026-tr.html) or the equipment manufacturer service information.
- Install the cooling fan, if equipped. See equipment manufacturer service information.
- Install the cooling fan drive belt. Refer to Procedure 008-002 in Section 8.
(/qs3/pubsys2/xml/en/procedures/216/216-008-002-tr.html)
- Install the accessory drive pulley belt, if equipped. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.

- Operate the engine and check for proper operation.

Last Modified: 14-Sep-2016
